

Python & C Programming

Task 03 — Loops, Patterns & Basic Automation

Complete Solutions with Sample Output

Part A: Multiplication Table Generator

Python Code

Python

```
# multiplication_table.py
n = int(input('Enter a number: '))
for i in range(1, 11):
    print(f'{n} x {i} = {n * i}')
```

OUTPUT

```
Enter a number: 7
7 x 1  = 7
7 x 2  = 14
7 x 3  = 21
7 x 4  = 28
7 x 5  = 35
7 x 6  = 42
7 x 7  = 49
7 x 8  = 56
7 x 9  = 63
7 x 10 = 70
```

C Code

C

```
// multiplication_table.c
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    for (int i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }
}
```

```
    return 0;
}
```

OUTPUT

```
Enter a number: 7
7 x 1  = 7
7 x 2  = 14
...
7 x 10 = 70
```

Part B: Number Analysis Tool

Python Code

Python

```
# number_analysis.py
n = int(input('Enter a number N: '))
total = n * (n + 1) // 2
evens = n // 2
odds  = n - evens
print(f'Sum          = {total}')
print(f'Even Numbers = {evens}')
print(f'Odd Numbers  = {odds}')
```

OUTPUT

```
Enter a number N: 10
Sum          = 55
Even Numbers = 5
Odd Numbers  = 5
```

C Code

C

```
// number_analysis.c
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number N: ");
    scanf("%d", &n);
    int sum  = n * (n + 1) / 2;
    int evens = n / 2;
```

```

    int odds  = n - evens;

    printf("Sum          = %d\n", sum);
    printf("Even Numbers = %d\n", evens);
    printf("Odd Numbers  = %d\n", odds);

    return 0;
}

```

OUTPUT

```

Enter a number N: 10

Sum          = 55
Even Numbers = 5
Odd Numbers  = 5

```

Part C: Pattern Printing Challenge

Python Code

Python

```

# patterns.py

# Pattern 1 - Increasing stars
print('Pattern 1:')
for i in range(1, 6):
    print('*' * i)

# Pattern 2 - Decreasing stars
print('\nPattern 2:')
for i in range(5, 0, -1):
    print('*' * i)

# Pattern 3 - Number triangle
print('\nPattern 3:')
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end='')
    print()

```

OUTPUT

```

Pattern 1:

*
**
***

```

```
****  
*****
```

Pattern 2:

```
*****  
****  
***  
**  
*
```

Pattern 3:

```
1  
12  
123  
1234  
12345
```

C Code

C

```
// patterns.c  
#include <stdio.h>  
  
int main() {  
    // Pattern 1  
    printf("Pattern 1:\n");  
    for (int i=1; i<=5; i++) {  
        for (int j=1; j<=i; j++) printf("*");  
        printf("\n");  
    }  
  
    // Pattern 2  
    printf("\nPattern 2:\n");  
    for (int i=5; i>=1; i--) {  
        for (int j=1; j<=i; j++) printf("*");  
        printf("\n");  
    }  
  
    // Pattern 3  
    printf("\nPattern 3:\n");  
    for (int i=1; i<=5; i++) {  
        for (int j=1; j<=i; j++) printf("%d", j);  
        printf("\n");  
    }  
}
```

```
}  
    return 0;  
}
```

OUTPUT

```
Pattern 1:  *  **  ***  ****  *****  
Pattern 2:  *****  *****  ***  **  *  
Pattern 3:  1  12  123  1234  12345
```

Part D: Password Attempt Simulator

Python Code

Python

```
# password_attempt.py  
password = 'secure123'  
attempts = 3  
  
for i in range(attempts):  
    guess = input(f'Enter password (Attempt {i+1}/{attempts}): ')  
    if guess == password:  
        print('Access Granted')  
        break  
else:  
    print('Account Locked')
```

OUTPUT

```
--- Scenario 1: Correct password on 2nd try ---  
Enter password (Attempt 1/3): hello  
Enter password (Attempt 2/3): secure123  
Access Granted  
  
--- Scenario 2: All attempts failed ---  
Enter password (Attempt 1/3): abc  
Enter password (Attempt 2/3): 1234  
Enter password (Attempt 3/3): wrong  
Account Locked
```

C Code

C

```
// password_attempt.c  
#include <stdio.h>
```

```
#include <string.h>

int main() {
    char password[] = "secure123";
    char guess[50];
    int attempts = 3, granted = 0;
    for (int i = 1; i <= attempts; i++) {
        printf("Enter password (Attempt %d/%d): ", i, attempts);
        scanf("%s", guess);
        if (strcmp(guess, password) == 0) {
            printf("Access Granted\n");
            granted = 1; break;
        }
    }
    if (!granted) printf("Account Locked\n");
    return 0;
}
```

OUTPUT

```
Enter password (Attempt 1/3): wrong
Enter password (Attempt 2/3): tryagain
Enter password (Attempt 3/3): nope
Account Locked
```

Part E: Mini Cyber Security Utility – Username Generator

Python Code

Python

```
# username_generator.py

first = input('Enter First Name: ').lower()
last = input('Enter Last Name: ').lower()
year = input('Enter Birth Year: ')
sy = year[2:]

print('\nSuggested Usernames:')
print(f'1. {first}{last}{year}')
print(f'2. {first[0]}.{last}{sy}')
print(f'3. {last}_{first}')
print(f'4. {first[:3]}{last[:3]}{sy}')
print(f'5. {first}.{last}{year}')
```

OUTPUT

Enter First Name: Soham

Enter Last Name : Patel

Enter Birth Year: 2004

Suggested Usernames:

1. sohampatel2004

2. s.patel04

3. patel_soham

4. sohpat04

5. soham.patel2004

C Code

C

```
// username_generator.c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void toLower(char *s) {
    for (int i = 0; s[i]; i++) s[i] = tolower(s[i]);
}

int main() {
    char first[50], last[50], year[10], sy[3];
    printf("Enter First Name: "); scanf("%s", first);
    printf("Enter Last Name : "); scanf("%s", last);
    printf("Enter Birth Year: "); scanf("%s", year);
    toLower(first); toLower(last);
    sy[0]=year[2]; sy[1]=year[3]; sy[2]='\0';
    printf("\nSuggested Usernames:\n");
    printf("1. %s%s%s\n", first, last, year);
    printf("2. %c.%s%s\n", first[0], last, sy);
    printf("3. %s_%s\n", last, first);
    printf("4. %.3s%.3s%s\n", first, last, sy);
    printf("5. %s.%s%s\n", first, last, year);
    return 0;
}
```

OUTPUT

Enter First Name: Soham

Enter Last Name : Patel

Enter Birth Year: 2004

Suggested Usernames:

1. sohampatel2004

2. s.patel04

3. patel_soham

4. sohpat04

5. soham.patel2004

Bonus: Number Guessing Game

Python Code

Python

```
# number_guessing.py
import random

secret = random.randint(1, 50)
attempts = 0
print('Guess a number between 1 and 50!')
while True:
    guess = int(input('Your guess: '))
    attempts += 1
    if guess < secret: print('Too low!')
    elif guess > secret: print('Too high!')
    else:
        print(f'Correct! You got it in {attempts} attempt(s).')
        break
```

OUTPUT

Guess a number between 1 and 50!

Your guess: 25

Too high!

Your guess: 12

Too low!

Your guess: 18

Too low!

Your guess: 21

Correct! You got it in 4 attempt(s).

C Code

C

```
// number_guessing.c
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    srand(time(0));

    int secret = rand() % 50 + 1;
    int guess, attempts = 0;
    printf("Guess a number between 1 and 50!\n");
    do {
        printf("Your guess: ");
        scanf("%d", &guess);
        attempts++;
        if (guess < secret) printf("Too low!\n");
        else if (guess > secret) printf("Too high!\n");
        else printf("Correct! You got it in %d attempt(s).\n", attempts);
    } while (guess != secret);
    return 0;
}
```

OUTPUT

```
Guess a number between 1 and 50!
Your guess: 10
Too low!
Your guess: 30
Too high!
Your guess: 20
Correct! You got it in 3 attempt(s).
```

Submitted by

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Python & C Programming — Task 03